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CYCLISTIC
BIKE-SHARE



ANALYSIS SUMMARY REPORT

How Do Annual Members & Casual Riders Use Cyclistic Bikes Differently?

A behavioral analysis of 5,842,831 rides to inform
Cyclistic's membership conversion strategy.

5.84M

Rides Analyzed

12 mo

Data Coverage

5

Key Findings

3

Recommendations

Sahan K.

Google DA Capstone | June 2026
sksahan911@gmail.com

01EXECUTIVE SUMMARY

The 30-second version

Cyclistic has two radically different user groups sharing one product. Annual members ride on weekdays during rush hour — they are commuters. Casual riders dominate weekends and ride 56% longer per trip — they are leisure users. Finance has confirmed members are significantly more profitable. This analysis of 5.84 million rides identifies when, where, and how to convert casuals into members.

64.1%

Annual Members
3,747,384 rides

35.9%

Casual Riders
2,095,447 rides

+56%

Duration Gap
Casuals ride longer

-93%

Casual Winter Drop
Aug to Jan decline

02DATA & METHODOLOGY

What we used and how

Data Source	Motivate International Inc. (Divvy) — Public License
Coverage	January 2025 – May 2026 (12 monthly CSV files)
Raw Volume	5,848,703 rows × 13 columns (before cleaning)
Clean Volume	5,842,831 rows (5,872 invalid rides removed = 0.10%)
Removed Rows	Duration < 60 sec (equipment checks), Duration > 24 hrs (lost bikes), missing station names
Key Columns	ride_id, rideable_type, started_at, ended_at, start/end station name & coords, member_casual
Derived Features	ride_duration_min, day_of_week, hour_of_day, month, season
Tools	Python 3.11 (pandas, numpy), openpyxl, pptxgenjs, reportlab

03KEY FINDINGS

5 data-backed behavioral insights

Ride Duration Gap

1

Casual average: 18.9 min (median 11.2). Member average: 12.1 min (median 8.6). The gap is most pronounced on weekends — casuals average 21.9 min on Sundays vs 13.2

+56%

Weekday vs. Weekend Split

2

Members peak Tuesday-Thursday (commute days). Casuals peak Saturday — 431,737 rides, representing 48% of all Saturday ridership. The behavioral split is so clean it

Sat Peak

Rush-Hour Commuter Pattern

3

Members generate 273,069 rides at 8am and 404,857 at 5pm — classic double-peak commuter behavior. Casuals show no morning spike: their rides build gradually from noon,

8am+5pm

4

Seasonal Volatility

Casual ridership drops 92.7% from peak (Aug 337K) to winter trough (Jan 25K). Members drop 75.0% over the same period. The spring re-entry window (April-June) is when casuals

-93% drop

5

Geographic Divide

Top casual stations: Navy Pier, Lake Shore Dr., Millennium Park, Shedd Aquarium — all tourist/leisure destinations. Top member stations: Canal St., Clinton St., State St. — all

0 overlap

04 RECOMMENDATIONS

Three actions directly tied to the data

01

Weekend Warrior Membership Tier

Casuals dominate Saturdays with 431K rides (48% of Saturday volume). A weekend-only membership priced between a day pass and full annual directly matches observed behavior. Activate at Navy Pier, Millennium Park, and Lake

Target: 1.1M weekend casual rides/yr | Channel: Station kiosks + app

02

Spring Conversion Campaign (Apr-Jun)

Casual ridership rebounds 900%+ from January to August. April–June is the re-entry window — riders are actively recommitting to cycling. Trigger a limited-time membership offer after a casual rider's 3rd ride of the season when

Window: April - June | Trigger: 3rd ride of season | Max intent period

03

In-App Cost Breakdown Feature

Casuals ride 56% longer per trip, meaning they pay more per mile on single-use passes than members. Show each casual: 'You spent \$X this season. A membership would save you \$Z.' Make the math personal after the 5th casual

Channel: In-app push | Trigger: 5th casual ride | Personalized ROI

05 LIMITATIONS & NEXT STEPS

What this data cannot answer

Limitation	Detail
No individual user IDs	Cannot track whether the same casual rider converts to a member. Behavioral patterns are aggregate.
No pricing data	Membership conversion ROI is estimated. Actual pricing elasticity needs finance input.
No demographic data	Cannot segment by age, income, or commute distance. All findings are behavioral.
Chicago-specific	All stations are in Chicago. Findings may not generalize to other Cyclicistic markets.
Partial calendar year	Jan 2026 data covers only 5 months. Full 2026 seasonality projection requires additional data.

Recommended Next Steps:

- Present to Lily Moreno and request campaign brief approval for all three recommendations.
- Model Weekend Warrior tier pricing with the finance team using member ARPU data.
- Scope the in-app cost comparison feature with the engineering team (estimated: 2-week build).
- Design the spring 2027 campaign for April re-entry targeting — the data-identified conversion window.

- Incorporate demographic data (if obtainable) to refine casual rider personas.

Analysis conducted under the Universal DA Checklist (10 phases, 76 checks). All critical checks passed: source documented, raw data preserved, cleaning logged, statistical claims include effect sizes, no causation inferred from correlation.